AZURITE AND MALACHITE

 Azurmalachite is a combination of copper carbonates azurite ($Cu_3(CO_3)_2(OH)_2$) and malachite ($Cu_2(CO_3)(OH)_2$).	**Characteristics:** Vivid blue and green bands formed by alternating azurite and malachite. Relatively soft with perfect cleavage. This makes it ideal for use in jewelry. **Natural Occurrences:** It develops at shallow depths in the oxidizing zones above copper deposits, where slightly acidic, copper-bearing waters percolate through fractures in the rock. As these waters descend, they create a geochemical environment conducive to carbonate precipitation, leading to the formation of azurmalachite where conditions alternate between those favorable for azurite and malachite.